

Information Systems

4IF - Business Intelligence

IDENTIFICATION

CODE : IF-4-BDD
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 0.0 h
Laboratory : 16.0 h
Project : 0.0 h
Teacher-student
contact : 25.0 h
Personal work : 25.0 h
Total : 50.0 h

ASSESSMENT METHOD

Final examen (1h30), documents on
paper can be used

TEACHING AIDS

Lecture slides

TEACHING LANGUAGE

French

CONTACT

MME TCHOUNIKINE Anne
anne.tchounikine@insa-lyon.fr
M. TOMMASINI Riccardo
riccardo.tommasini@insa-lyon.fr

AIMS

The lecture objective is the learning of the models, methods and tools needed for the design and the implementation of BI projects.

Acquired skills are, being able to:

- Identify report objects and indicators for a BI project
- Design multidimensional models
- Contribute to the design and implementation of dashboards
- Elaborate OLAP requests
- Be able to operate in BI environments

CONTENT

The course syllabus will address the 4 following main topics:

- Decisional information systems and multidimensional modelisation,
- Design and implementation of data warehouses,
- Information analysis based on a data warehouse [OLAP and reporting]
- Management of BI projects.

BIBLIOGRAPHY

- A. Vaisman, E. Zimanyi. Data Warehouse Systems, Design and Implementation. Springer-Verlag
M. Golfarelli, S. Rizzi. Data Warehouse Design: Modern Principles and Methodologies. McGraw-Hill Osborne
C. Webb, A. Ferrari, M. Russo. Expert Cube Development with SSAS Multidimensional Models. Packt Publishing
G. Spofford. MDX Solutions. Wiley.
W. H. Inmon. Building the Data Warehouse. Wiley.
J. Gray et al. Data Cube: A Relational Aggregation Operator Generalizing Group-By, Cross-Tab, and Sub-Totals. Data Mining and Knowledge Discovery, March 1997
R. Kimball, M. Ross. The Data Warehouse Toolkit, 3rd Edition. Wiley.

PRE-REQUISITE

Relational database